# Software Architecture and Design Specification

Project: Smart Expense & Budget Management Application

Version: 1.0

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Date: 05-10-2026

Status: Draft for review

## Revision History

*Table 1 – Revision history*

| Version | Date | Author | Change Summary |
|---|---|---|---|
| 1.0 | 05-10-2026 | Team 6 | Initial SASD. |

## Approvals

*Table 2 – Approvals*

| Role | Name | Signature / Date |
|---|---|---|
| Team Member | Shriram Chandrasekar (PES2UG24CS495) | |
| Team Member | Shreya S Mattimadu (PES2UG24CS486) | |
| Team Member | Shashwath K R (PES2UG24CS463) | |
| Team Member | Shashank A (PES2UG24CS461) | |
| Product Owner (per STP v1.0) | Sudeepa Roy Dey | |

## 1. Introduction

### 1.1 Purpose

This document specifies the architecture and design of v1.0 of the application: pattern, components, technology and data stores, security, and the mapping from SRS v1.0 requirements to design. It contains no source code and no test cases.

### 1.2 Scope

In scope: SRS v1.0 in full – users (FR1–FR3), expenses (FR4–FR7), categories (FR8–FR9), budgets (FR10–FR12), analytics (FR13–FR15), alerts (FR16–FR18), reports (FR19–FR20), audit and backup (FR21–FR22), NFR1–NFR14.

Out of scope: moving money, and live bank connectivity or bank-account integration of any kind. FR7 is satisfied by the user uploading a statement file; the application does not connect to a bank. Also out of scope: native mobile apps (NFR7); push notifications (FR18, optional); trained ML models, since categorisation is rule-based (FR9).

### 1.3 Audience

Developers, the project guide and product owner, the test team, the evaluator and maintainers.

### 1.4 Definitions

FR / NFR – requirement as numbered in SRS v1.0. SRS / STP / RTM – Software Requirements Specification v1.0 (05-09-2026), Software Test Plan, Requirements Traceability Matrix. MVC – Model–View–Controller. API – JSON over HTTPS/TLS. Utilisation – percentage of a budget limit spent in its period (FR11). Backup – a restorable copy of stored data (FR22).

# 2. Document Overview

### 2.1 How to use this document

Section 3 is the architectural baseline that constrains implementation and records how SRS v1.0 requirements are addressed. Section 4 is the build-level contract: decisions, sequence diagrams, API, error handling, UX and open issues. Requirement IDs are quoted inline; “shall” is binding. Where the SRS leaves a decision open, this document states the design intent and records the decision as an open issue rather than presenting it as settled.

### 2.2 Related Documents

Project Proposal and F&NFR document (2026); SRS_team 6 (v1.0, 05-09-2026); Software Test Plan (v1.0); Use Case Diagram (v1.0); data dictionary and RTM – to be produced.

# 3. Architecture

### 3.1 Goals & Constraints

Goals, each traceable to a non-functional requirement:

- Modularity – features change independently (NFR11).
- Security – encryption at rest and in transit, hashed passwords, expiring tokens, no SQL injection, XSS or CSRF (NFR4–NFR6).
- Performance – dashboard and analytics in 2–3 s for 10,000 transactions per user; reports in 5 s for one year (NFR1, NFR2).
- Usability – one responsive UI for mobile, tablet and desktop, WCAG basics (NFR7, NFR8).
- Reliability and privacy – 99% uptime target, daily backups, audit trail, no third-party sharing (NFR9, NFR10, NFR14).

Constraints, which limit the design space:

- C1 – stack not frozen: Python (Django) or JavaScript (MERN) is open (OI-1). No decision below depends on the framework.
- C2 – modular MVC is mandatory (NFR11, SRS 2.5); browser-based, JSON over HTTPS/TLS.
- C3 – store may be relational or document-based, but must hold users, expenses, categories and budgets (SRS 3.2); only sandbox libraries (SRS 2.6; STP 11).
- C4 – four developers, one semester, no operations team; operating complexity is a design concern.

### 3.2 Stakeholders & Concerns

- User – privacy of financial records, correct totals and remaining balance, speed, no duplicate alerts.
- Shared-budget admin and members – clear admin separation, audited changes, no access to another member's data (FR3, SRS 6.5).
- Development and test teams – low coupling, explicit interfaces, fixtures for the 10,000-transaction performance case.
- Guide, evaluator and external providers – visible requirement coverage and risks, and isolation of third-party failure so that it cannot affect stored financial data.

## 3.3 Component (UML) Diagram

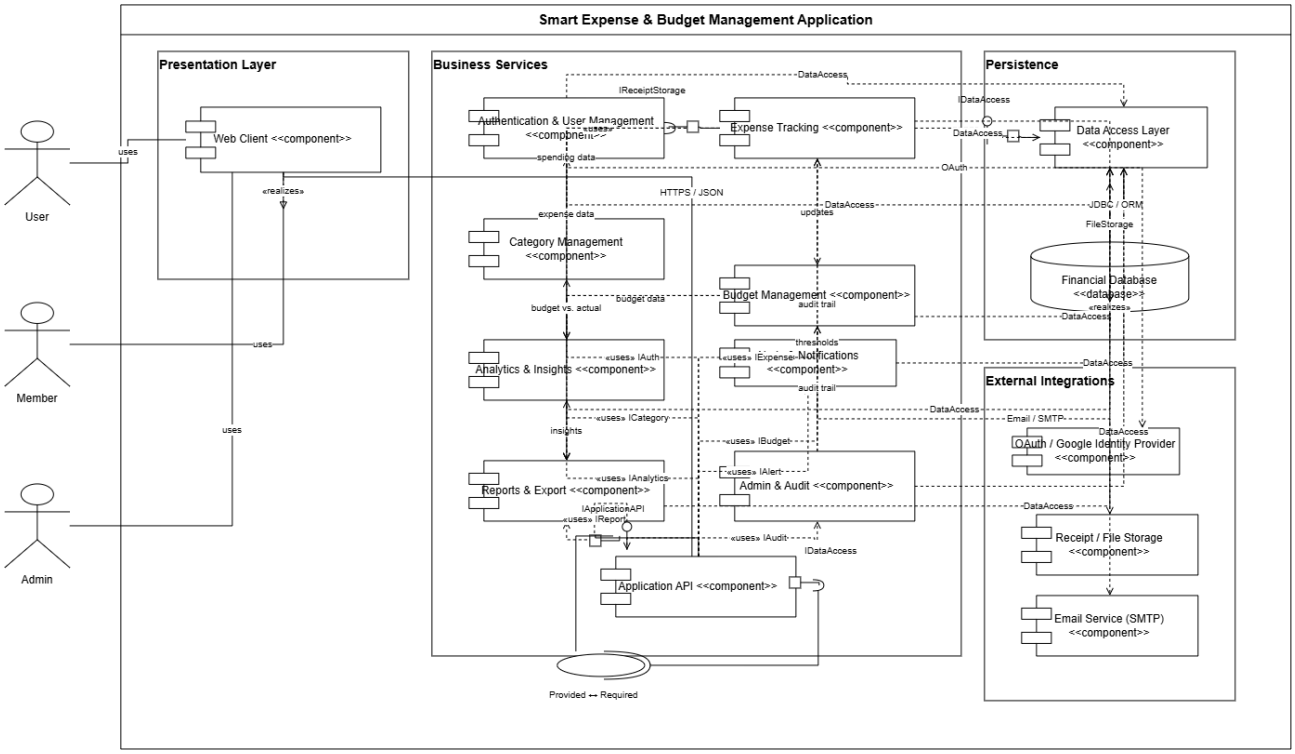



*Figure 3.1 – Component diagram. Responsibilities in Table 3; PlantUML source is held with the project files.*

## 3.4 Component Descriptions

*Table 3 – Component descriptions*

| Component | Responsibility (requirements) |
|---|---|
| Web Client | Responsive UI; input validation for feedback; no business rules (NFR7, NFR8). |
| REST API | Terminates HTTPS; authn/authz, validation, error envelope; never touches a store (NFR4–NFR6). |
| Auth & User | Register, log in, optional OAuth/Google, bcrypt, credential issue and invalidation (FR1–FR3; mechanism per OI-2). |
| Expense | Expense CRUD with soft delete, receipts, recurring rules, statement and CSV import (FR4–FR7). |
| Category | Default and custom categories with sub-categories; advisory suggestion (FR8, FR9). |
| Budget | Budgets per period and category; utilisation, remaining, rollover or reset (FR10–FR12). |
| Analytics | Aggregates by category, period, payment mode; series, projection, comparison (FR13–FR15). |
| Alert | Threshold and anomaly checks; one alert per budget, threshold, period; in-app, e-mail (FR16–FR18). |
| Report | Range validation, PDF/Excel via an export adapter, stored artefact (FR19, FR20). |
| Audit & Backup | Audit entry written with every business change; scheduled backup and restore (FR21, FR22). |
| Scheduler | Recurring expenses, period close, nightly backup; every job idempotent (FR6, FR12, NFR10). |
| Relational store | System of record for business data; the primary backup target (FR22). |
| File store | Receipts and generated report artefacts; recoverable, outside the web root (FR5, NFR4). |
| External services | Transactional e-mail behind an adapter, isolated from the committing write; OAuth/Google optional (FR18, FR1). |

## 3.5 Chosen Architecture Pattern and Rationale

A layered three-tier architecture implementing MVC, deployed as a modular monolith: presentation layer (Web Client), API layer (one JSON API over HTTPS/TLS), application layer (one module per capability, each with its own controllers, services and data access), data layer (relational and file store). Modules are developed, unit-tested and replaced independently.

- Satisfies the mandatory modular MVC constraint (NFR11) and is proportionate to four developers and no operations team.
- The application tier is designed to support horizontal scaling (NFR3). The authentication and session mechanism is still open (OI-2); it will be chosen with its shared-state requirements accounted for, so statelessness is a design target rather than a property of the current design.
- Microservices rejected – distributed transactions (an expense write also updates utilisation and may raise an alert, Figure 4.2) and per-service deployment; no goal above requires them. The boundaries still allow Analytics and Report to be extracted later.
- A broker, a primary document store or a separate analytics store rejected – the domain needs uniqueness, referential integrity and exact decimals. Indexes and database-side aggregation are planned to support the 10,000-transaction target (NFR1), to be validated by performance testing (TC-Perf-01).
- Layering rules: the client never reaches a store; cross-cutting concerns are implemented once; every write is atomic with its audit entry (FR21).

## 3.6 Technology Stack & Data Stores

*Table 4 – Technology stack and data stores*

| Concern | Decision for v1.0 |
|---|---|
| Language, framework | Python (Django) or JavaScript (Node/Express) – open, OI-1. |
| Server, client | REST controllers plus a service layer per capability; responsive HTML/CSS with a JavaScript UI layer (NFR7, NFR11). |
| Interface | JSON over HTTPS/TLS only; ISO-8601 dates; amounts carried and stored as exact decimals, never floating point (FR11, FR20). |
| Security | Token-based authentication (JWT or server-side session, OI-2) with expiry and invalidation; bcrypt; ownership and role checks per request (NFR4, NFR5). |
| Primary store | One relational database for all business data: User, Category, Expense, Budget, Alert, SharedBudget, AuditEntry (SRS 3.2). |
| Binary storage | File or object store outside the web root; authorised short-lived downloads only (NFR4). |
| Libraries | Charting and PDF/Excel behind adapter interfaces; transactional e-mail (C3). |
| Scheduling, logs | Platform scheduler for recurring expenses, period close, nightly backup, all idempotent; application and error logs with a request id, rotated (NFR10, NFR12). |
| Deploy, secrets | One application container, store as a managed service or second container; credentials from environment configuration only (NFR13). |

Backup scope (FR22, NFR10): the relational database is backed up daily. Receipt files referenced by user records are included in the backup set, or held in a store from which they can be recovered, since a receipt cannot be reconstructed from the database alone. Report artefacts may be regenerated and are not part of the record. Restore is to be rehearsed in the test environment (R6). The file store holds no business rows and is served only through a controlled endpoint (NFR4). No separate analytics store in v1.0.

## 3.7 Risks & Mitigations

*Table 5 – Risks and mitigations*

| Risk | Mitigation | Owner |
|---|---|---|
| R1 – stack not frozen, so framework-specific work cannot start and a late decision forces rework. | Decide after this review (OI-1); then the ORM, session mechanism and deploy target follow. | Team 6 |
| R2 – duplicate or missed alerts when expenses post together or a period closes. | Uniqueness on (budget, threshold, period); idempotent sweep; dispatch decoupled from the write. | Shashwath K R |
| R3 – response times exceed NFR1 or NFR2. | Indexes on (user, date) and (user, category); aggregation in the database; pagination; one-year report cap (OI-3); to be validated by TC-Perf-01. | Shashank A |

| Risk | Mitigation | Owner |
|---|---|---|
| R4 – a third-party library is unavailable or differs in the sandbox. | External capabilities behind adapters (C3); versions pinned; stubs kept ready. | Shashwath K R |
| R5 – sensitive data exposed, or a member acts as shared-budget admin. | TLS; bcrypt; credential expiry and invalidation (OI-2); ownership and role checks; nothing sensitive logged; SRS 6.5 enforced in Budget. | Shriram Chandrasekar |
| R6 – backups taken but never proven, and a restore fails. | Nightly backup of the database and receipt set; restore rehearsed in the test environment; outcome is a monitored signal (4.4). | Shashwath K R |

## 3.8 Traceability to Requirements

A test identifier appears only where the STP already establishes it: FR1, FR4, FR10, FR16 and NFR1. All other rows are planned verification; the full RTM is maintained separately.

*Table 6 – Traceability to requirements*

| Requirements | Design element | Verification |
|---|---|---|
| FR1 – register, log in, optional Google sign-in | Auth & User (3.4); Figure 4.1; POST /auth/login (4.3) | TC-Auth-01, TC-Auth-02 |
| FR2, FR3 – profile; shared budget and roles | Auth & User, Budget (3.4); PATCH /users/me, POST /shared-budgets (4.3) | RTM / planned |
| FR4 – add, edit, view, delete expenses | Expense module (3.4); soft delete (D4); Figure 4.2 | TC-Exp-01, TC-Exp-02 |
| FR5–FR7 – receipts; recurring; import, CSV or statement file | Expense, File store, Scheduler (3.4, D3, D6); import job, per-row errors (4.3, 4.4) | RTM / planned |
| FR8, FR9 – categories; suggestion | Category module; rule-based, advisory (3.4, D8) | RTM / planned |
| FR10 – create budgets per category | Budget module (3.4); budget endpoints (4.3) | TC-Bud-01 |
| FR11, FR12 – utilisation; rollover or reset | Budget module (3.4); period dates, close job (D2); utilisation endpoint (4.3) | RTM / planned |
| FR13–FR15 – charts; projection; comparison | Analytics module, charting adapter (3.4, 3.6); analytics endpoints (4.3) | RTM / planned |
| FR16 – threshold alerts | Alert module, uniqueness (3.4, D5); Figure 4.2 | TC-Alert-01 |
| FR17, FR18 – anomaly and reminder alerts | Alert module; anomaly rule open (OI-6); channel preferences (4.3) | RTM / planned |
| FR19–FR22 – reports and export; audit; backup and restore | Report module and export adapters (3.4); audit entry in the same unit of work (3.5); nightly backup of the database and receipt set (3.6, R6) | RTM / planned |
| NFR1 – performance, 10,000 transactions | Indexes and database-side aggregation (3.5, 3.6); R3 | TC-Perf-01 |
| NFR2, NFR3 – one-year reports; scaling | Date-range cap (OI-3); tier designed for horizontal scaling, session mechanism open (3.5, OI-2) | RTM / planned |
| NFR4–NFR6 – encryption; token expiry; injection, XSS, CSRF | TLS, bcrypt (3.6); Figure 4.1; threat model and explicit NFR6 controls (3.9) | STP 5 security cases; RTM for the remainder |
| NFR7, NFR8 – responsive UI; accessibility | Responsive layout and screen inventory (4.5) | STP 5 usability cases; RTM for the remainder |
| NFR9–NFR14 – uptime; backup; modular MVC; logging; container; privacy | Health signal (4.4, OI-5); nightly backup (3.6); layered MVC (3.5); shared logging (4.1, 4.4); container (3.6); no third-party data flows (3.9) | RTM / planned |

## 3.9 Security Architecture

Applied uniformly by the API layer and the modules, not per feature. Passwords are stored only as salted bcrypt hashes and never logged (NFR4). Credentials expire and can be invalidated server-side; the mechanism is JWT or server-side session and is recorded as OI-2, and all traffic is HTTPS only (NFR5). Expense, budget and report records are personal financial data: encrypted in transit and at rest, readable only by the owner or shared-budget members, audited on every change, never sent to a third party (FR21, NFR14). Receipts sit outside the web root behind authorised short-lived endpoints (FR5).

*Table 7 – Threat model (STRIDE)*

| Threat | Design mitigation | Req. |
|---|---|---|
| Spoofing | bcrypt hashing; credentials with expiry, verified and invalidated server-side; ownership and role checks per request. | NFR4, NFR5 |
| Tampering | Client never trusted: server-side re-validation; ORM or parameterised statements only; bad import rows rejected individually; every change audited. | NFR6, FR21 |
| Repudiation | Append-only audit entries: actor, entity, action, timestamp, request id, before/after; expenses soft-deleted so history survives. | FR21 |
| Information disclosure | File store outside the web root; error envelope carries a code and safe message only; TLS throughout. | NFR4, NFR6 |
| Denial of service | Upload size and type limits; row and date-range caps; pagination; rate limiting on import, report and alert endpoints. | NFR1, NFR2 |
| Elevation of privilege | Default deny: membership and shared-budget settings exist only on admin-only endpoints, re-checked in Budget; role read from the membership record. | FR3 |

NFR6 coverage is explicit: server-side validation with ORM or parameterised statements mitigates SQL injection; output encoding and safe template rendering mitigate XSS; CSRF protection is applied where cookie-based or session authentication is selected, and is not required for a bearer-token scheme (OI-2). Password reset and recovery are absent from SRS v1.0 and are recorded as OI-4. Payment threats are excluded: no card or bank credentials are stored and no money moves.

# 4. Design

## 4.1 Design Overview

- D1 – amounts are stored and computed as exact decimals, never floating point, so totals, balances and reports agree to the last minor unit (FR11, FR20).
- D2 – a budget stores explicit period start and end dates, so utilisation is unambiguous (FR10–FR12).
- D3 – a recurring expense is a rule materialised into an ordinary expense when due, so history needs no special case (FR6).
- D4 – expenses are soft-deleted: they leave lists and utilisation but stay for audit and past reports (FR21).
- D5 – one alert per budget, threshold and period, so retries, imports and sweeps cannot duplicate (FR16, FR18; R2).
- D6 – a receipt is a file and the row keeps a reference, served only through an authorised endpoint (FR5, NFR4).
- D7 – timestamps stored in UTC, budget periods evaluated in the user's time zone, avoiding off-by-one-day errors.
- D8 – category suggestion is advisory: the user confirms or overrides, and the choice becomes history (FR9).

Cross-cutting concerns are implemented once, in the API layer or a shared utility layer: authentication and authorisation on every request, re-checked for ownership and role inside the owning module; shape validation at the API layer, business rules in the module; output encoding at the render layer; one error envelope; one request id in every log line and audit entry; the audit entry written in the same unit of work as the change.

## 4.2 UML Sequence Diagrams

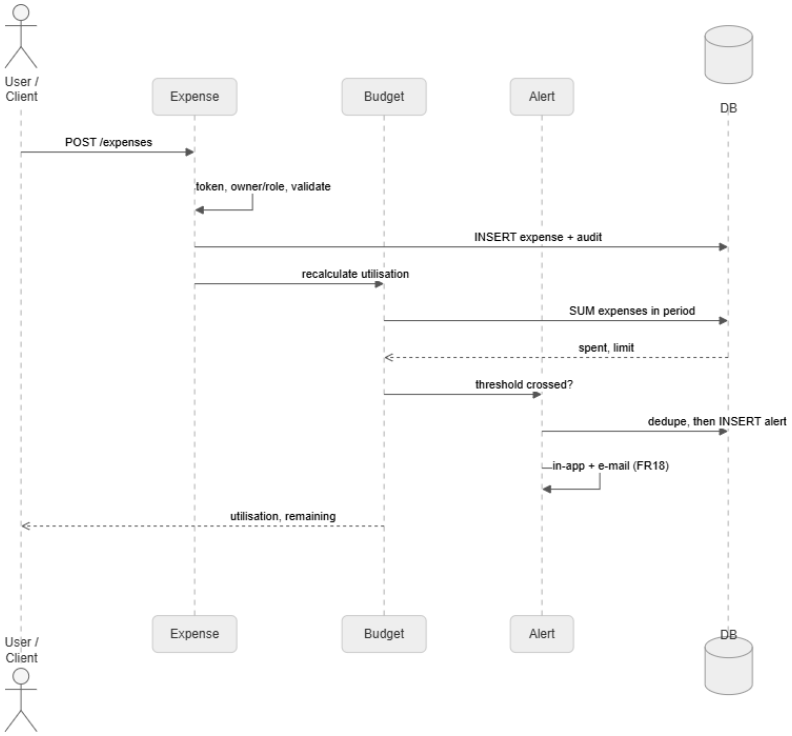



*Figure 4.1 – Authentication and session establishment (FR1–FR3, NFR4, NFR5): both FR1 sign-in paths, bcrypt verification, the non-revealing 401, and credential issue with expiry.*

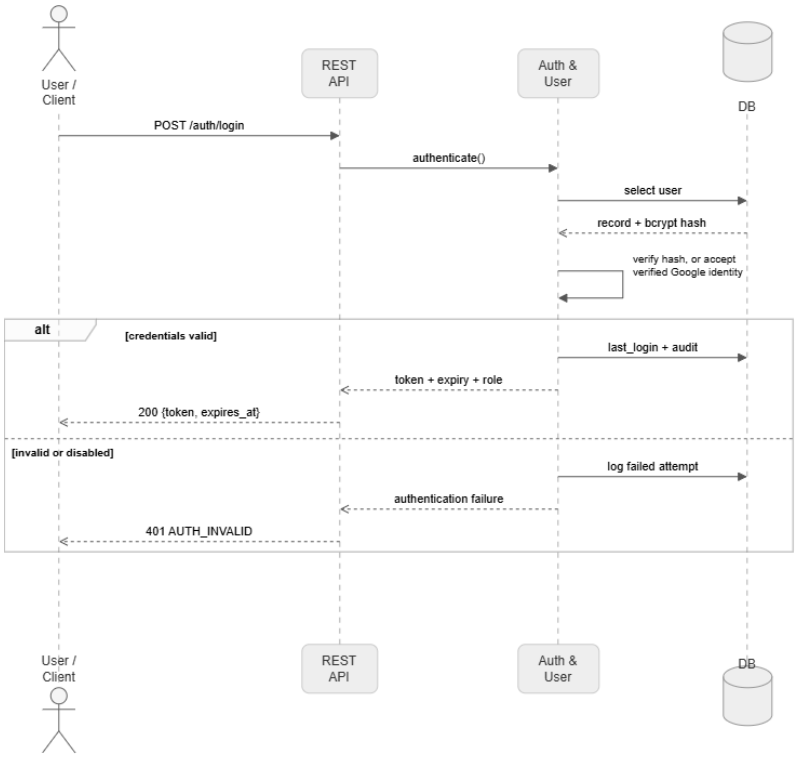


*Figure 4.2 – Recording an expense, updating utilisation and raising a threshold alert (FR4, FR9, FR11, FR16, FR18, FR21). The alert is de-duplicated before dispatch, and notification failure cannot fail the write.*

Report generation (FR19, FR20), statement and CSV import (FR7), period close (FR12) and the nightly backup (FR22) use the same participants; their diagrams are deferred (OI-5).

## 4.3 API Design

JSON over HTTPS/TLS only (C2). Conventions for every endpoint in Table 8 – base path /api/v1; Authorization header carrying the credential except on registration and login, verified with expiry and invalidation before routing (NFR5); identity taken from the presented credential only; ISO-8601 dates; pagination with a server-enforced maximum; 201 create, 200 update, 204 delete, 202 plus a job resource for long work; the error envelope of 4.4 on every failure.

*Table 8 – API endpoints*

| Module | Endpoint | Purpose (authorisation) |
|---|---|---|
| Auth & User | POST /auth/register, /auth/login, /auth/logout; GET, PATCH /users/me | Create account; obtain or invalidate a credential; own profile (public / authenticated). |
| Expense | GET, POST /expenses; GET, PATCH, DELETE /expenses/{id}; /expenses/{id}/receipt; /expenses/import; /recurring-expenses | List, create, update, soft-delete; attach receipt; import CSV or bank statement file as a job; recurring rules (authenticated + owner). |
| Category | GET, POST, PATCH, DELETE /categories; GET /categories/suggest | Manage categories; advisory suggestion, FR8–FR9 (authenticated). |
| Budget | GET, POST /budgets; PATCH, DELETE /budgets/{id}; GET /budgets/{id}/utilisation | Create, amend, remove; spent, limit, remaining, percent used (authenticated + owner). |
| Alert | GET /alerts; POST /alerts/{id}/read; GET, PUT /alert-preferences | List and mark read; thresholds and channels (authenticated). |
| Analytics | GET /analytics/summary, /by-category, /trend, /projection, /comparison | Totals, series, projection, comparison, FR13–FR15 (authenticated). |
| Report; Audit; Shared | POST /reports; GET /reports/{id}/download; GET /audit; POST /shared-budgets | Report and download; audit trail; shared budget, FR3 (owner / admin). |

```
POST /api/v1/expenses        GET /api/v1/budgets/bdg_301/utilisation
{ "amount": "1250.50",      { "limit_amount": "10000.00",
  "date": "2026-10-04",      " "spent_amount": "9670.85", "percent_used": 96.7,
  "category_id": "cat_17"    " "thresholds_crossed": [80], "alert_raised": true }

201 { "id": "exp_55021",
      "budget_impact": { "budget_id": "bdg_301" } }
```

Expense: amount positive; the caller owns the record or belongs to the shared budget charged; write and audit entry atomic (FR21). Budget: utilisation aggregates that period's expenses for the category, excluding soft-deleted rows (D4). Crossing a budget threshold raises an alert; it does not reject the expense. The SRS requires alerts at 80% and 100% (FR16) and does not require blocking, so the expense is recorded and overspend is reported separately. A rejection rule is not designed for v1.0.

Import (FR7): the user uploads a CSV or bank statement file; accepted columns and the statement formats supported are to be confirmed (OI-7), as no bank format has been selected. Rows are validated individually and the job returns accepted and rejected counts with per-row errors. The application does not connect to a bank.

## 4.4 Error Handling, Logging & Monitoring

Modules throw a typed error with a stable code; the API layer adds a safe message and the request id, so errors read the same whichever module raised them: { "error": { "code": "PERIOD_CLOSED", "message": "The October budget period is closed and no longer accepts expenses.", "field": "date", "request_id": "req_7c1f" } }. Stack traces, SQL fragments and file paths go to the error log, never to the client.

*Table 9 – Error codes*

| HTTP | Example codes | Client behaviour |
|---|---|---|
| 400 | VALIDATION_ERROR, INVALID_AMOUNT | Highlight the field; do not retry unchanged. |
| 401 / 403 | AUTH_INVALID, CREDENTIAL_EXPIRED, NOT_OWNER | Return to sign-in; refuse without revealing whether the record exists. |

| HTTP | Example codes | Client behaviour |
|---|---|---|
| 404 / 409 | EXPENSE_NOT_FOUND; EMAIL_EXISTS, PERIOD_CLOSED | Refresh the list; the user resolves the conflict. |
| 413 / 415 / 422 | PAYLOAD_TOO_LARGE, UNSUPPORTED_MEDIA_TYPE, IMPORT_ROWS_REJECTED | Reject the file, state accepted types, show per-row errors. |
| 429 / 500 / 503 | RATE_LIMITED, INTERNAL_ERROR, EMAIL_UNAVAILABLE | Back off; generic message quoting the request id. A notification failure never rolls back a change already committed. |

- Logging – one application line per significant event plus a separate error log for unhandled and 5xx conditions, both rotated; each line carries timestamp, level, request id, user id where authenticated, module, operation, outcome and duration. Never logged: passwords, hashes, credentials, receipt contents, card or bank details. The audit trail is a business record, append-only.
- Monitoring – signals a team member can act on: sign-in page and health endpoint availability (NFR9 is a target; how it is measured is open, OI-5); 5xx rate; expenses, budget-summary and analytics above 3 s (NFR1); report job above 5 s (NFR2); nightly backup outcome (NFR10); failed e-mail dispatches. A production observability stack is out of scope for v1.0.

## 4.5 UX Design

One responsive web interface for mobile, tablet and desktop (NFR7). The screens are those SRS 3.1 lists plus those FR2 and FR3 require; Table 10 maps each to its requirement.

*Table 10 – Screens*

| Screen | Key elements | Req. |
|---|---|---|
| Sign in / Register | E-mail and password, show/hide password, error text beside the field, “sign in with Google” if enabled. | FR1 |
| Dashboard | Budget progress bars with spent, limit, remaining; period totals; latest expenses; unread alert count; month-end projection. | FR10, FR11, FR14 |
| Expenses, entry, import | Filterable, sortable list; entry form with amount, date, payment mode, merchant, notes, category with advisory suggestion, receipt upload; import picker for CSV or statement file with per-row results. | FR4, FR5, FR7, FR9 |
| Recurring; Categories; Budgets | Rule list with cadence and next run date; defaults read-only, custom categories; per-category limits, period, rollover or reset, alert thresholds. | FR6, FR8, FR10–FR12, FR16 |
| Analytics; Alerts; Reports | Category pie or bar, trend line, comparison, projection; unread badge, alert list with reason and link to the record; date range with a one-year cap, format, download link. | FR13–FR20 |
| Settings; Shared budget | Profile and account settings; the admin creates the shared budget and manages members and roles, members add and view shared expenses. | FR2, FR3 |

- One responsive set of screens across phone, tablet and desktop (NFR7); no separate mobile app.
- Beyond NFR8: fields visibly labelled, adequate contrast, focus order follows the visual order, controls keyboard operable, charts carry a text alternative or data table, errors shown beside the field.
- Formatting and degradation: amounts always shown with their unit, dates in the user's locale, overspend distinguished by more than colour; destructive actions confirm and state the effect on related data; if e-mail fails the expense is still saved and the user is told.

## 4.6 Open Issues & Next Steps

OI-1 blocks stack-specific implementation as a whole; the other items block or constrain only the feature areas named in them, and must be resolved before those areas are finalised.

*Table 11 – Open issues*

| ID | Open issue | Needed by / Owner |
|---|---|---|
| OI-1 | Freeze the stack: Python (Django) or JavaScript (MERN). Framework, ORM and deploy target follow from it (R1). | Immediate / Team 6 |
| OI-2 | Choose JWT or server-side sessions; fix credential lifetime and invalidation. Sections 3.5, 3.9 and 4.3 state the requirement and are written to hold either choice. | With OI-1 / Shriram Chandrasekar |
| OI-3 | Confirm the maximum report and chart date range and the message beyond it (NFR2 assumes one year). | Before Reports / Shashank A |

| ID | Open issue | Needed by / Owner |
|---|---|---|
| OI-4 | Password reset, recovery and e-mail verification are not in SRS v1.0; decide which optional features enter v1.0 (OAuth/Google, push, ML). | Next revision / Team 6 |
| OI-5 | Add the remaining sequence diagrams; produce the physical schema and data dictionary; agree retention for audit entries, receipts and reports, and how 99% uptime is measured. | Before Analytics and Reports / Team 6 |
| OI-6 | Define the FR17 anomaly rule: multiple of the user's category average, minimum sample size, false-positive tolerance. | Before anomaly alerts / Shashwath K R |
| OI-7 | Confirm the accepted import column layout and which bank statement file formats FR7 must support; no bank format has been selected. | Before import / Shashank A |

Next steps: record the stack decision (OI-1) as an ADR; set up repository, environment and container per the Project Proposal; then implement in dependency order – Auth & User, Expense and Category, Budget, audit write infrastructure (FR21, built alongside the first module rather than after it), Alert, Analytics, Report, and finally Backup and Restore (FR22).

# 5. Appendices

### 5.1 Glossary

*Table 12 – Glossary*

| Term | Definition |
|---|---|
| ADR | Architecture decision record: a significant design decision and its rationale, in writing. |
| Audit trail | Append-only record of who changed which expense, budget or report, when (FR21). |
| Soft delete | Flagging a record as deleted so it leaves user views but is retained for audit and past reports (D4). |

### 5.2 References

SRS v1.0 (05-09-2026), F&NFR Requirements, Project Proposal, Software Test Plan and Use Case Diagram – Team 6, 2026. IEEE Std 42010. OWASP Top 10 and ASVS (threat model, STP 5.1). WCAG 2.1 (NFR8).

### 5.3 Tools

draw.io for UML diagrams, the stack's charting and PDF/Excel libraries behind adapters; Docker; Git; STP test tooling. An OpenAPI description of 4.3 follows the stack decision (OI-1).